Non-classical Modal Logics

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Available at `www.leonardopacheco.xyz/slides/aizu.pdf`

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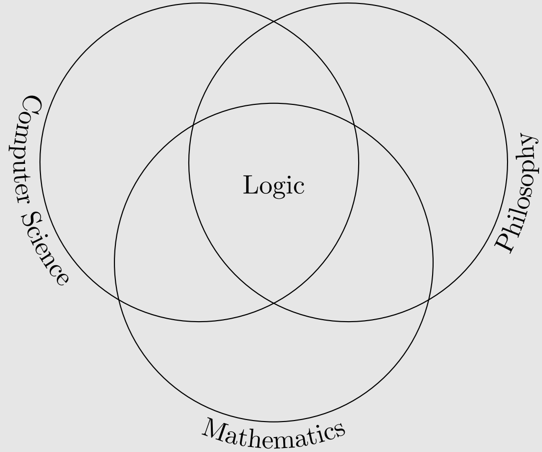
Career overview — Education

- 2012–2018: Universidade Federal de Goiás, Brazil
  - Two years as undergraduate in Computer Science
  - One year as Exchange Student at Tohoku University
  - Undergraduate in Mathematics
- 2018–2023: Tohoku University, Japan
  - M.Sc. in Mathematics, supervised by Kazuyuki Tanaka
  - Ph.D. in Mathematics, supervised by Keita Yokoyama
  - TA for Mathematical Logic and Computability courses
  - Kawai Doctoral Thesis Award

Career overview — Research

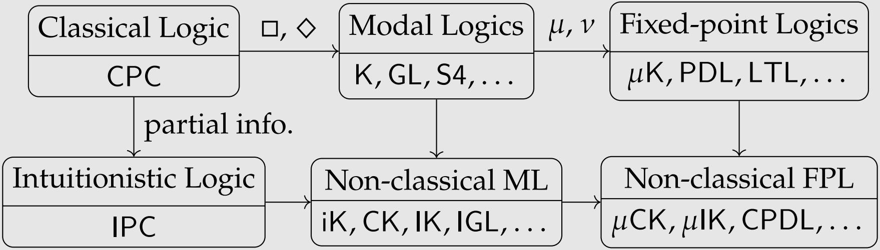
- 2023–2024: TU Wien, Austria
  - Postdoctoral researcher
  - Supervised by Juan P. Aguilera
- 2024–present: Institute of Science Tokyo, Japan
  - JSPS Postdoctoral Fellow
  - KAKENHI Grant 24KF0214:
    "Constructive μ-calculus for model checking"
  - Supervised by Ryo Kashima
- Reviewing:
  - Fuzzy Sets and Systems, Logic Journal of the IGPL, Annals of Pure and Applied Logic, IJCAR 2026, AiML 2026, LICS 2026, TABLEAUX 2025
  - Program committee of 8th Intuitionistic Modal Logic and Applications Workshop

Logic is here:



My research area

I work with the combination of modal logics, intuitionistic logic, and fixed-point logics.



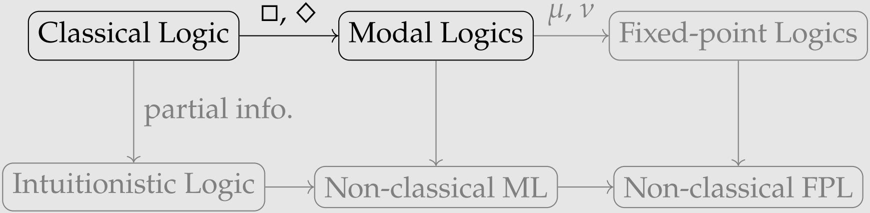
Objective

Logics to reason about programs with partial information.

Some history

- Modal Logics
  - Modern semantics [Kri59]
  - Reasoning about programs [FL77]
- Fixed-point Logics:
  - μ-calculus [Koz83]
  - Decidability [Wal00]
- Intuitionistic Logic
  - Instuitionism [Bro23]
  - A logic of partial information [Vic89]
- Non-classical Modal Logic
  - Some scattered old works [Ono77, Sim94]
  - Surge of research since [DM23]

Modal logics



Modal logics

Modal logics add to propositional logic:

- $\Box$: necessity;
- $\Diamond$: possibility.

These allow us to state *how* statements are true.

Example

Three interpretations of $\Box\varphi$*:*

- φ *is provable;*
- φ *holds after the execution of the program* `p`*;*
- *the agent a knows that* φ.

Models for modal logic

A model for modal logic is:

$$M = \langle W, R, V \rangle,$$

where:

- $\langle W, R \rangle$ is a directed graph
- $V : \mathrm{Prop} \to \mathcal{P}(W)$ specifies the states where each proposition is true.

Example

$$\begin{tikzcd}
 & u \in V(Q) \\
w \arrow[ur] \arrow[dr] & \\
 & v \in V(P) \arrow[uu]
\end{tikzcd}$$

Interpreting formulas

Let $M = \langle W, R, V \rangle$ and $w \in W$, then

- $M, w \models \Box\varphi$ if, for all v, if wRv then $M, v \models \varphi$;
- $M, w \models \Diamond\varphi$ if there is v such that wRv and $M, v \models \varphi$.

Example

$$\begin{tikzcd}
 & u \in V(Q) \\
w \arrow[ur] \arrow[dr] & \\
 & v \in V(P) \arrow[uu]
\end{tikzcd}$$

- $M, w \models \Box(P \vee Q)$
- $M, v \not\models \Diamond P$

The modal logic K

We define the modal logic K as the set of formulas valid over all models:

Definition

Let φ be a formula.
Then $\mathsf{K} \models \varphi$ if, for every model $M = \langle W, R, V\rangle$ and $w \in W$,

$$M, w \models \varphi.$$

Proof systems

We can also define logics via proof systems.

Example

A proof of K *is a finite labeled tree:*

$$\dfrac{\dfrac{\dfrac{\dfrac{\varphi \vdash \varphi, \psi \qquad \varphi, \psi \vdash \psi}{\varphi \to \psi, \varphi \vdash \psi}}{\Box(\varphi \to \psi), \Box\varphi \vdash \Box\varphi}}{\Box(\varphi \to \psi) \vdash \Box\varphi \to \Box\psi}}{\vdash \Box(\varphi \to \psi) \to (\Box\varphi \to \Box\psi)}$$

The semantics and proof-theoretic definitions of K coincide:

Theorem (Soundness and Completeness)

For every formula φ, $\mathsf{K} \models \varphi$ *if and only if* $\mathsf{K} \vdash \varphi$.

Other modal logics

We can obtain other logics by modifying our semantics or our proofs:

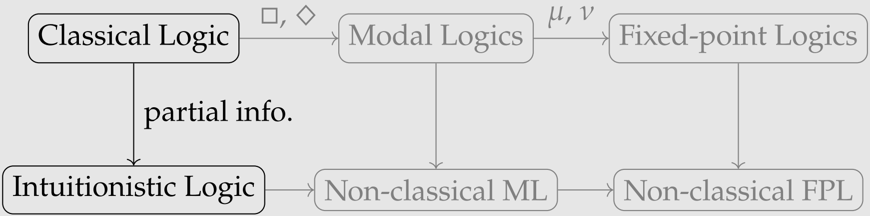
Example

K4 *is the logic obtained by*

- *requiring R to be transitive; or*
- *adding the rule*

$$\frac{\Gamma, \Box\Gamma \vdash \varphi}{\Box\Gamma \vdash \Box\varphi}$$

Intuitionistic propositional logic



Intuitionistic propositional logic

In intuitionistic logic, we cannot use:

- excluded middle: $\varphi \vee \neg\varphi$,
- double negation exclusion: $\neg\neg\varphi \to \varphi$.

More precisely, proof carries evidence:

- To prove $\varphi \vee \psi$, we prove one of φ or ψ and identify which one;
- To prove $\varphi \to \psi$, we define a construction that converts proofs of φ into proofs of ψ.

Intuitionistic logic and partial information

A model for intuitionistic propositional logic is:

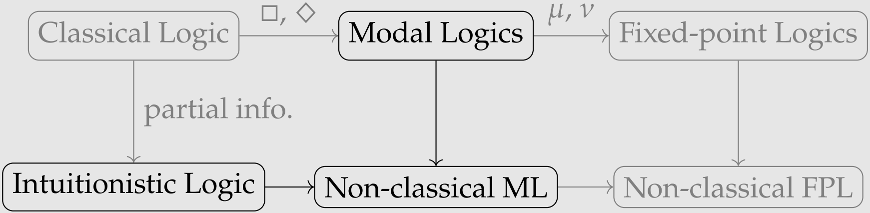
$$M = \langle W, \leq, V \rangle,$$

where:

- $\langle W, \leq \rangle$ is a directed graph;
- $\leq$ is a reflexive and transitive relation;
- $V : \mathrm{Prop} \to \mathcal{P}(W)$ satisfies

$$w \leq v \text{ and } w \in V(P) \text{ imply } v \in V(P).$$

Intuitionistic propositional logic



Non-classical modal logic

A model for the non-classical modal logic CK is

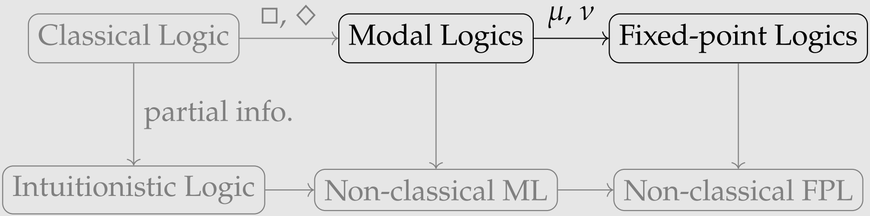
$$M = \langle W, \leq, R, V \rangle,$$

where:

- $\langle W, R, V \rangle$ is a model for modal logic, and
- $\langle W, \leq, V \rangle$ is a model for intuitionistic propositional logic.

There are choices for the interpretations of $\Box$s and $\Diamond$s, we do not discuss these in this presentation.

Fixed-point logics



Fixed-point logics

We can express "infinite" properties by finite formulas if we use fixed-points:

Example

Interpret $\Box\varphi$ *as "*φ *holds after the execution of a program* $\mathtt{p}$*", then*

$$\nu X.(\varphi \land \Box X) \approx \varphi \land \Box\varphi \land \Box\Box\varphi \land \cdots ,$$

that is,

$$\nu X.(\varphi \land \Box X) \approx \text{“}\varphi \textit{ always holds”}.$$

The μ-calculus is a modal logic containing both least and greatest fixed-points. It balances expressivity and computational complexity:

- validity is decidable in EXPTIME-complete,
- model checking is polynomial time, in general.

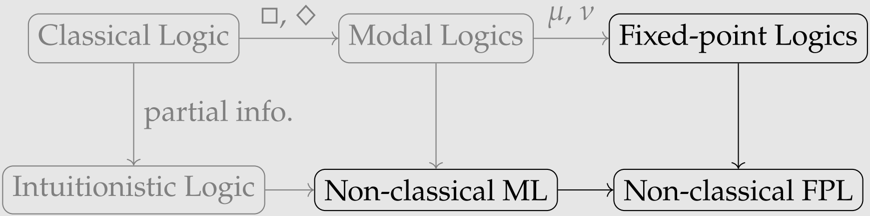
Fixed-points, semantically

Let M be a model, then $\nu X.(\varphi \wedge \Box X)$ is the greatest fixed-point of the operator

$$\begin{aligned}\Gamma : \mathcal{P}(W) &\to \mathcal{P}(W)\\ A &\mapsto \{w \mid M[X \mapsto A], w \models \varphi \wedge \Box X\},\end{aligned}$$

where $M[X \mapsto A]$ is the same as M, except $V(X) = A$.

Non-classical fixed-point logics



Non-wellfounded proof systems

Another way to get new logics is to allow infinite proofs:

$$
\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\vdots}{\Box(\Box P \to P) \vdash P, \Box P} \qquad \Box(\Box P \to P), P \vdash P}{\Box(\Box P \to P), \Box P \to P \vdash P}}{\Box(\Box P \to P) \vdash P, \Box P} \qquad \Box(\Box P \to P), P \vdash P}{\Box(\Box P \to P), \Box P \to P \vdash P}}{\Box(\Box P \to P) \vdash \Box P}}{\vdash \Box(\Box P \to P) \to \Box P}
$$

GL proves $\Box(\Box P \to P) \to \Box P$.

Cyclic proof systems

Better yet, use cyclic proofs to encode infinite proofs:
—todo: write loop—

$$
\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\Box(\Box P \to P) \vdash P, \Box P} \qquad \Box(\Box P \to P), P \vdash P}{\Box(\Box P \to P), \Box P \to P \vdash P}}{\Box(\Box P \to P) \vdash P, \Box P} \qquad \Box(\Box P \to P), P \vdash P}{\Box(\Box P \to P), \Box P \to P \vdash P}}{\Box(\Box P \to P) \vdash \Box P}}{\vdash \Box(\Box P \to P) \to \Box P}
$$

GL proves $\Box(\Box P \to P) \to \Box P$.

Intuitionistic Gödel–Löb IGL

Theorem (Das, van der Giessen, Marin [DvdGM24])

Assume Σ^1_1-determinancy. The non-wellfounded proof system $m\ell\mathsf{IGL}$ is sound and complete with respect to IGL-models.

Note: Σ^1_1-determinancy is not provable over ZFC.

Theorem (Aguilera, Pacheco [AP25])

The cyclic proof-system $cm\ell\mathsf{IGL}$ is sound and complete with respect to IGL-models.

- No strong assumption.
- IGL is semi-decidable.
- First work combining labeled and cyclic proof systems.

Constructive μ-calculus

We may evaluate $M, w \models \varphi$ by a game between two players: Verifier and Refuter.

Theorem (Pacheco [Pac26], preprint)

- *Relational semantics and game semantics for the constructive μ-calculus are equivalent.*
- *The non-wellfounded proof system* $\mathsf{lab}\mu\mathsf{CK}_\omega$ *is sound and complete with respect to the constructive μ-calculus.*

- First work on the full μ-calculus in a non-classical setting.
- Extends to other non-classical μ-calculi.
- Non-wellfounded proofs as universal strategies for valid formulas.

Topological semantics

Theorem (McKinsey–Tarski [MT44])

S4 *is complete with respect to topological spaces if we interpret* $\Box$ *as the interior operator.*

Theorem (Montacute, Pacheco [MP], submitted)

iK4, iGL, iS4, iGrz, iCK4, IEL, PLL, mHC, iSL, *and* KM *are complete with respect to their topological models.*

Theorem (Aguilera, Fernández-Duque, Pacheco [AFDP26], preprint)

CK4, IK4, GK4, CS4, IS4, *and* GS4 *are complete with respect to their topological models.*

TOPOLOGIES, □S AND ◇S

- iK4 has only □s.
- iK4-models need only one topology:

$$\langle W, \tau_{\Box}, V \rangle.$$

- IK4 has both □s and ◇s.
- IK4-models need three topologies:

$$\langle W, \tau_{\to}, \tau_{\Box}, \tau_{\Diamond}, V \rangle.$$

Intuitionistic epistemic logic

The logic IEL [AP16] formalizes:

intuitionistic truth implies intuitionistic knowledge.

Theorem (Pacheco, Sedlár [PS], in preparation)

The following two views are compatible with IEL*:*

- *knowledge is necessity* ($\Box$)*;*
- *knowledge is possibility* ($\Diamond$)*.*

Other results

Expressivity over various classes of models:

- Pacheco, "The μ-calculus' Alternation Hierarchy is Strict over Non-Trivial Fusion Logics", 2025.
- Pacheco, Tanaka, "The alternation hierarchy of the mu-calculus over weakly transitive frames", 2022.

Connection between computation, fixed-points, large ordinals, and infinite games:

- Aguilera, Lubarsky, Pacheco, "Higher-Order Feedback Computation", 2024.

Reverse Mathematics of infinite games and reflection principles:

- Pacheco, Yokoyama, "Determinacy and reflection principles in second-order arithmetic", arXiv:2209.04082.

Research Plans

- Short-term: basic theory of non-classical logics.
  - decidability of IGL.
  - cyclic proof system and decidability of non-classical μ-calculi.
- Mid-term: application of the basic theory.
  - Extend the above to logics such as PDL and LTL.
  - Model checking: Does a finite-state model satisfy a given specification?
  - Knowledge representation: Represent information in a structured, machine-processable form.
- Long-term: other non-classical logics.
  - Paraconsistent logic: allows contradictory information.
  - Relevant logic: stronger correlation required for implication.

Future Prospects

- Better logics to reason about programs and partial information.
- Many problems in non-classical modal logics still open.
- Clarify the relation between $\Box$ and $\Diamond$.

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Juan P. Aguilera, David Fernández-Duque, and Leonardo Pacheco, *Polytopological semantics for intuitionistic modal logics*, 2026.

Sergei Artemov and Tudor Protopopescu, *Intuitionistic epistemic logic*, Rev. Symb. Log. **9** (2016), no. 2, 266–298. MR 3506909

Juan P. Aguilera and Leonardo Pacheco, *Intuitionistic Gödel-Löb without sharps*, ACM Trans. Comput. Log. **26** (2025), no. 4, Art. 20, 14. MR 4998554

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- Anupam Das and Sonia Marin, *On Intuitionistic Diamonds (and Lack Thereof)*, Automated Reasoning with Analytic Tableaux and Related Methods (Revantha Ramanayake and Josef Urban, eds.), Lecture Notes in Computer Science, Springer Nature Switzerland, 2023, pp. 283–301.

- Anupam Das, Iris van der Giessen, and Sonia Marin, *Intuitionistic Gödel-Löb Logic, à la Simpson: Labelled Systems and Birelational Semantics*, 22:1–22:18.

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- Dexter Kozen, *Results on the propositional μ-calculus*, Theoretical computer science **27** (1983), no. 3, 333–354.
- Saul A Kripke, *A completeness theorem in modal logic*, The Journal of Symbolic Logic **24** (1959), no. 1, 1–14.
- Yoàv Montacute and Leonardo Pacheco, *Coderivative intuitionistic logics of topological and bimetric spaces*, submitted.
- J. C. C. McKinsey and Alfred Tarski, *The algebra of topology*, Annals of Mathematics **45** (1944), no. 1, 141–191.

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- Hiroakira Ono, *On some intuitionistic modal logics*, Publ. Res. Inst. Math. Sci. **13** (1977), no. 3, 687–722. MR 476407
- Leonardo Pacheco, *The constructive μ-calculus: Game semantics and non-wellfounded proof systems*, 2026.
- Leonardo Pacheco and Igor Sedlár, *Intuitionistic knowledge as a constructive diamond*, in preparation.
- Alex K. Simpson, *The proof theory and semantics of intuitionistic modal logic*, Ph.D. thesis, University of Edinburgh, 1994.
- Steven Vickers, *Topology via logic*, Cambridge University Press, 1989.

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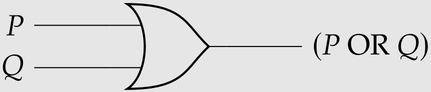
Igor Walukiewicz, *Completeness of kozen's axiomatisation of the propositional μ-calculus*, Information and Computation **157** (2000), no. 1-2, 142–182.

模擬講義：命題論理の意味論

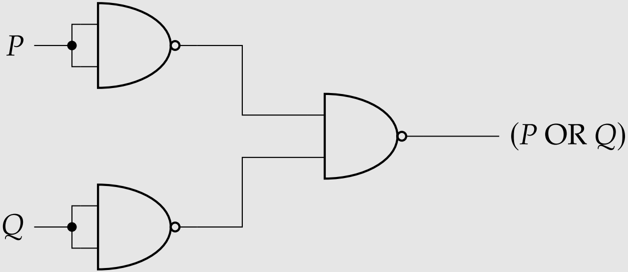
- 命題とは
- 意味論：割り当てと真理値表
- 同値
- 演算子の完全性

完全性と論理回路

論理回路 OR を以下で表す：



OR は NAND だけで表せる：



命題とは

真 (T) か偽 (F) かが定まる主張を命題 (proposition) と呼ぶ。

例

- 「８月４日に会津若松市で雪が降ります」は偽である。
- 「任意の実数 x に対し、$\sqrt{x}$ は実数である」は偽である。
- 「$2 + 2 == 4$」は真である。

命題を形式的に表すため、命題記号の集合

$$\mathrm{Prop} = \{P, Q, R, S, \ldots\}$$

を固定する。

論理式

演算子 $\neg$、$\wedge$、$\vee$、$\to$ を使って、論理式を定義する：

定義

- 命題記号は論理式である。
- φ と ψ が論理式ならば、

$$(\neg\varphi)、(\varphi \wedge \psi)、(\varphi \vee \psi)、(\varphi \to \psi)$$

  はいずれも論理式である。

以上に該当する記号列以外は論理式ではない。

論理式の例

例

- $(\neg P) \to Q$ は論理式である。
- $(\to P) \to Q$ は論理式でない。

(外側の括弧を省略する。)

真理値割り当て

定義

$V : \mathrm{Prop} \to \{T, F\}$ は命題記号に対する真理値割り当て *(valuation)* である。

例

$V(P) =, V(Q) = F, V(R) = F, \ldots$

命題記号に対する割り当て V が与えられたときに、V を任意の論理式に拡張することができる。

「でない」の真理値表

V を任意の論理式に拡張するために、以下のような真理値表 (truth table) を使う：

P	$\neg P$
T	
F	

「でない」の真理値表

V を任意の論理式に拡張するために、以下のような真理値表 (truth table) を使う：

P	$\neg P$
T	F
F	T

例

$V(P) = T$ ならば、$V(\neg P) = F$ である。

「かつ」の真理値表

P	Q	$P \land Q$
T	T	
T	F	
F	T	
F	F	

「かつ」の真理値表

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

例

$V(P) = F$ かつ $V(Q) = T$ ならば、$V(P \wedge Q) = F$ である。

「または」の真理値表

P	Q	$P \vee Q$
T	T	
T	F	
F	T	
F	F	

「または」の真理値表

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

例

$V(P) = F$ かつ $V(Q) = T$ ならば、$V(P \vee Q) = T$ である。

「ならば」の真理値表

P	Q	$P \to Q$
T	T	
T	F	
F	T	
F	F	

「ならば」の真理値表

P	Q	$P \to Q$
T	T	T
T	F	F
F	T	T
F	F	T

例

$V(P) = T$ かつ $V(Q) = F$ ならば、$V(P \to Q) = F$ である。

例：$(\neg P) \to Q$ の真理値表

P	Q	$\neg P$	$(\neg P) \to Q$
T	T		
T	F		
F	T		
F	F		

例：$(\neg P) \to Q$ の真理値表

P	Q	$\neg P$	$(\neg P) \to Q$
T	T	F	T
T	F	F	T
F	T	T	T
F	F	T	F

例：$(P \wedge Q) \wedge (R \wedge S)$ の真理値表

P	Q	R	S	$P \wedge Q$	$R \wedge S$	$(P \wedge Q) \wedge (R \wedge S)$
T	T	T	T	T	T	T
T	T	T	F	T	F	F
T	T	F	T	T	F	F
T	T	F	F	T	F	F
T	F	T	T	F	T	F
T	F	T	F	F	F	F
T	F	F	T	F	F	F
T	F	F	F	F	F	F
F	T	T	T	F	T	F
F	T	T	F	F	F	F
F	T	F	T	F	F	F
F	T	F	F	F	F	F
F	F	T	T	F	T	F
F	F	T	F	F	F	F
F	F	F	T	F	F	F
F	F	F	F	F	F	F

同値

定義

φ と ψ を論理式とする。任意の真理値割り当て V に対し $V(\varphi) = V(\psi)$ が成り立つとき、φ と ψ は同値 *(equivalent)* である。

例

$(\neg P) \to Q$ と $P \vee Q$ が同値である：

P	Q	$\neg P$	$(\neg P) \to Q$	$P \vee Q$
T	T	F	T	T
T	F	F	T	T
F	T	T	T	T
F	F	T	F	F

演算子の完全性

定義

X を演算子の集合とする。任意の論理式 φ に対し X の演算子のみを用いた論理式 ψ が存在して φ と ψ が同値になるとき、X を完全 *(functionally complete)* と呼ぶ。

例

- $\{\neg, \rightarrow\}$ は完全である。
- $\{\rightarrow\}$ は完全でない。

$\lor$ と $\land$ を $\neg$ と $\to$ で表せる

P	Q	$\neg P$	$(\neg P) \to Q$	$P \lor Q$
T	T	F	T	T
T	F	F	T	T
F	T	T	T	T
F	F	T	F	F

P	Q	$\neg Q$	$P \to (\neg Q)$	$\neg(P \to (\neg Q))$	$P \land Q$
T	T	F	F	T	T
T	F	T	T	F	F
F	T	F	T	F	F
F	F	T	T	F	F

$\{\neg, \rightarrow\}$ は完全である

論理式に表す $\vee$ と $\wedge$ を書き直せば、

定理

$\{\neg, \rightarrow\}$ は完全である。

例

$P \vee (Q \wedge R)$ と $(\neg P) \rightarrow (\neg(Q \rightarrow (\neg R)))$ は同値である。

$\{\to\}$ は完全でない

P	$P \to P$
T	T
F	T

φ が P と $\to$ のみを用いる論理式とする。

- $V(P) = T$ ならば、$V(\varphi) = T$ である。
- $V(\varphi) = V(\neg P)$ は成り立たない。

定理

$\{\to\}$ は完全でない。

まとめ

- 命題論理の論理式と意味論を定義した。
- 意味論に基づいて、同値を導入した。
- 演算子の完全性を調べてみた。

次に演習問題に取り組む。

演習問題 1

問題

演算子 $NAND$ を $|$ と書く。$NAND$ の真理値表は：

P	Q	$P \mid Q$
T	T	F
T	F	T
F	T	T
F	F	T

演算子の集合 $\{|\}$ が完全であることを証明せよ。

目的：

- 真理値表の理解の確認。
- 演算子の完全性の理解の確認。

演習問題 2

問題

演算子 XOR を $\nleftrightarrow$ と書く。XOR の真理値表は：

P	Q	$P \nleftrightarrow Q$
T	T	F
T	F	T
F	T	T
F	F	F

演算子の集合 $\{\nleftrightarrow\}$ が完全であることを証明せよ。

目的：

- さらに意味論と演算子の完全性の理解を深める。

Thank you!
ありがとうございます！

Slides available at `www.leonardopacheco.xyz/slides/aizu.pdf`

演習問題 1 ：回答

P	$P \mid P$	$\neg P$
T	F	F
F	T	T

P	Q	$Q \mid Q$	$P \mid (Q \mid Q)$	$P \to Q$
T	T	F	T	T
T	F	T	F	F
F	T	F	T	T
F	F	T	T	T

演習問題２：回答

P	$P \nleftrightarrow P$
T	F
F	F

φ が P と $\nleftrightarrow$ のみを用いる論理式とする。

- $V(P) = F$ ならば、$V(\varphi)$ である。
- $V(\varphi) = V(\neg P)$ は成り立たない。

定理

$\{\nleftrightarrow\}$ は完全でない。